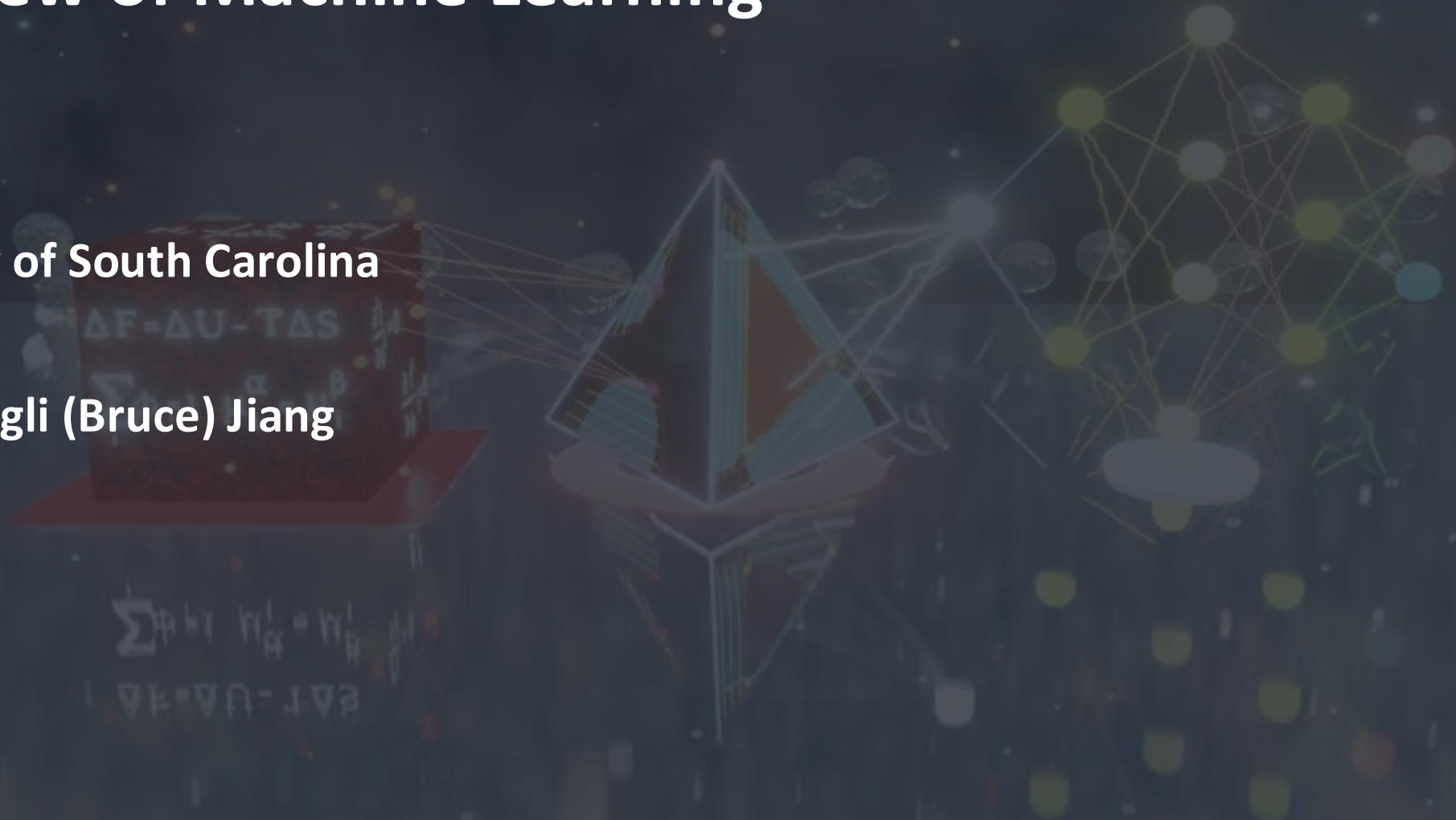


Overview of Machine Learning

Fall 2026

University of South Carolina

Prof. Shengli (Bruce) Jiang



Overview of Machine Learning

From Lectures 2 and 3

Machine learning involves interrelated tasks of

- *representing data,*
- *modeling data, and*
- *learning from data*

We have now reviewed mathematics of

- *linear algebra and analytical geometry*

and Python basics of

- *NumPy, SciPy, pandas, and matplotlib*

*Concepts, Supervised Learning, Unsupervised Learning,
Regression, Classification*

By the end of class, you should be able to *identify the sample, feature $\mathbf{X}$, label y , task, model, and prediction in a scientific ML problem.*

The Essential Elements of (most) ML

To use most ML methods, we will need to conceptualize our problem into the following:

Features (inputs, descriptors)

$$\{x_i\} \leftrightarrow X$$

x_i *feature vector of sample i*

- a numerical description of (ideally) characteristics that distinguish one sample from another
- may (or may not) have direct implications on the modeling outputs

<i>Features X</i>				
Reactor run	Temperature (K)	Pressure (bar)	Feed fraction	Conversion
1	350	4.0	0.30	0.72
2	370	4.5	0.35	0.81
3	390	5.0	0.40	0.88

Sample

The Essential Elements of (most) ML

To use most ML methods, we will need to conceptualize our problem into the following:

Labels (outputs)

$$\{y_i\} \leftrightarrow \mathbf{y}; \{\mathbf{y}_i\} \leftrightarrow \mathbf{Y}$$

y_i OR $\mathbf{y}_i$ *scalar or vector
label of sample i*

- usually a numerical (integer or real) description of sample i
- usually reserved for some special quantity or property of interest

Reactor run	Temperature (K)	Pressure (bar)	Feed fraction	Conversion
1	350	4.0	0.30	0.72
2	370	4.5	0.35	0.81
3	390	5.0	0.40	0.88

Labels $\mathbf{y}$

The Essential Elements of (most) ML

To use most ML methods, we will need to conceptualize our problem into the following:

Model

a function that operates on features

$$f(\mathbf{x}) \text{ or } \hat{f}(\mathbf{x})$$

- *often defines a mapping from feature space to label space*

Features X				Conversion
Reactor run	Temperature (K)	Pressure (bar)	Feed fraction	
1	350	4.0	0.30	0.72
2	370	4.5	0.35	0.81
3	390	5.0	0.40	0.88

Sample

Labels y

Predictions

the function output or predicted labels

$$\hat{y} = f(\mathbf{x}) \text{ or } \hat{y} = \hat{f}(\mathbf{x})$$

$$X \rightarrow f(\mathbf{x}) \rightarrow \hat{y}$$

- y is the measured target (**true** value)
- $\hat{y}$ is the model prediction (**predicted** value)

The Essential Elements of (most) ML

To use most ML methods, we will need to conceptualize our problem into the following:

Labeled Data

$$\{(\boldsymbol{x}, y)_i\}$$

*a set of tuples where
features and labels
are known*

Unlabeled Data

$$\{\boldsymbol{x}_i\}$$

*labels are not
necessarily known or
provided with features*

***What is an example of
unlabeled data?***

Features X				
Reactor run	Temperature (K)	Pressure (bar)	Feed fraction	Conversion
1	350	4.0	0.30	0.72
2	370	4.5	0.35	0.81
3	390	5.0	0.40	0.88

Sample

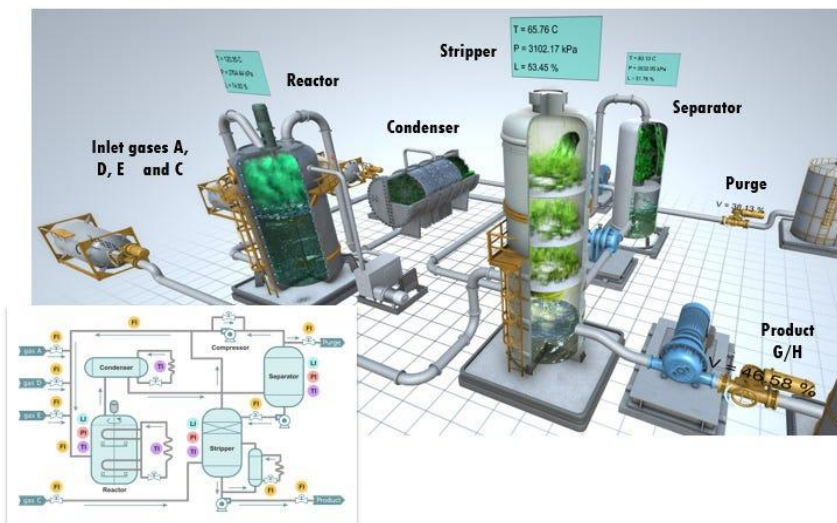
Labels y

$$X \rightarrow f(\boldsymbol{x}) \rightarrow \hat{y}$$

- y is the measured target (**true** value)
- $\hat{y}$ is the model prediction (**predicted** value)

A quick practice

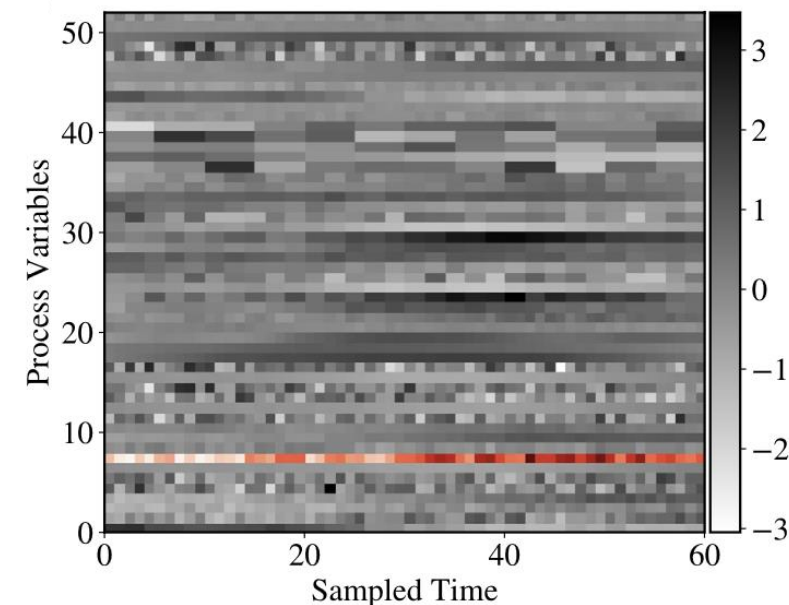
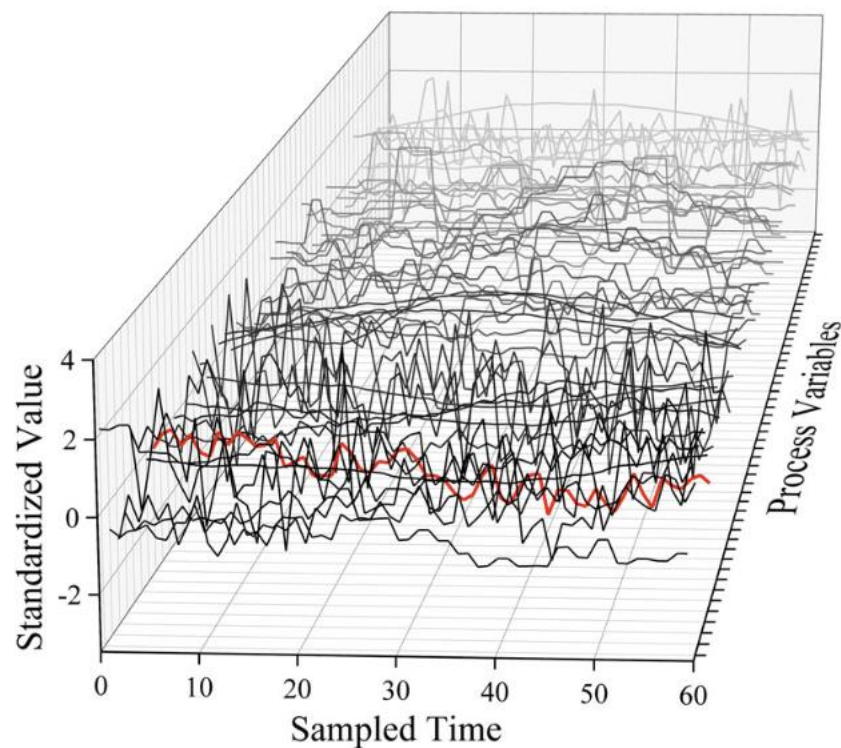
Tennessee Eastman Process



TEP dataset is simulated multivariate times-series data from realistic continuous chemical plant.

It contains process **measurements** and **control variables** under normal operation and different faults, such as feed changes, cooling-water problems, and equipment failures.

How do I represent this sample?



What can be a label?

The Essential Elements of (most) ML

Features (inputs, descriptors)

$$\{\mathbf{x}_i\} \leftrightarrow \mathbf{X}$$

$\mathbf{x}_i$ feature vector of sample i

Labels (outputs)

$$\{y_i\} \leftrightarrow \mathbf{y}; \{\mathbf{y}_i\} \leftrightarrow \mathbf{Y}$$

y_i OR $\mathbf{y}_i$ scalar or vector
label of sample i

Labeled Data

$$\{(\mathbf{x}, y)_i\}$$

a set of tuples where
features and labels
are known

Model

a function that operates on features

$$f(\mathbf{x}) \text{ or } \hat{f}(\mathbf{x})$$

- often defines a mapping from
feature space to label space

Predictions

the function output or
predicted labels

$$\hat{y} = f(\mathbf{x}) \text{ or } \hat{\mathbf{y}} = \hat{f}(\mathbf{x})$$

Unlabeled Data

$$\{\mathbf{x}_i\}$$

labels are not
necessarily known or
provided with features

Classes of Machine Learning

Supervised Learning

- In ***supervised learning***, we aim to create a model that can predict ***y*** as a function of ***x***.
- The optimization/learning of our model is ***supervised*** because the algorithm will exploit knowledge of labels over the dataset.

Supervised learning can be used for either

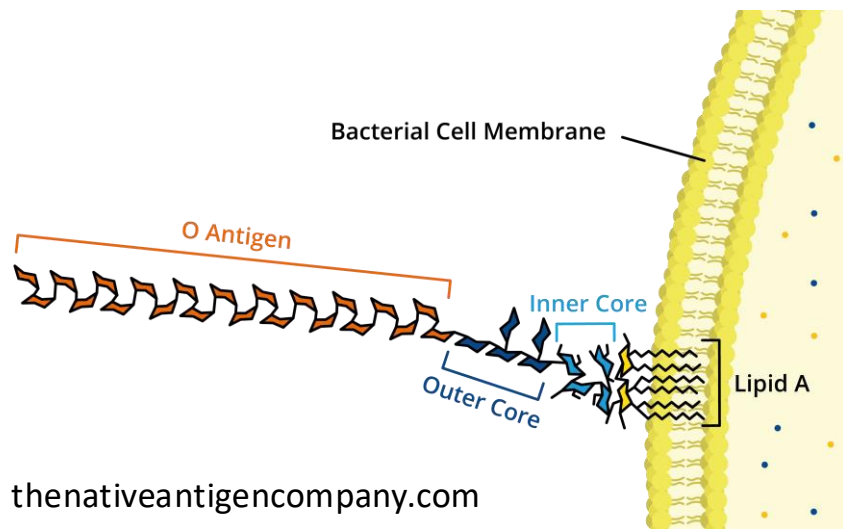
Regression – predict a *continuous* label. This is likely to be true for QSPR problems in physical science.

e.g., conductivity, melting point, band gaps

Classification – predict *categorical* labels or class membership. This can be useful for characterizing discrete outcomes.

e.g., (in)soluble, (un)synthesizable, (in)activity, hazardous

Regression Example on Bioanalytical Chemistry



Pharmaceutical Manufacturing

Medical Device Testing

Water Testing

Food and Beverage Testing



nhm.ac.uk

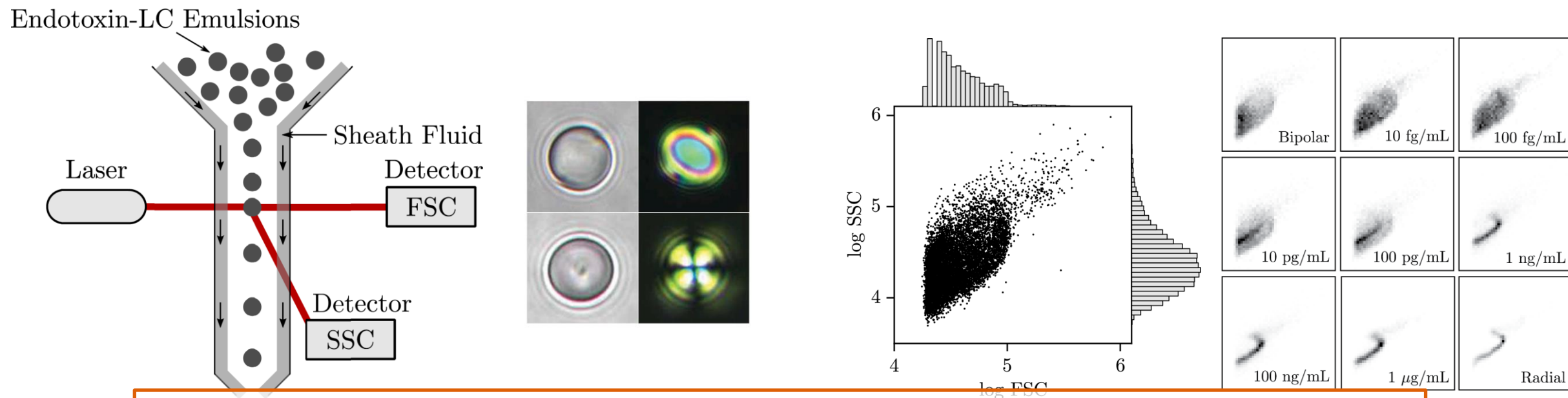
- **Endotoxins** are components of the outer membrane of gram-negative bacteria such as *E. coli*.
- They are released when bacterial cells die.
- In the bloodstream, **lipid A** triggers immune responses that can cause fever and septic shock.

Motivation

- The LAL test is widely used to detect endotoxins.
- It relies on horseshoe crab blood and can detect concentrations as low as 3.125 pg/mL.
- Need a new method to predict concentration and bacterial species.

Horseshoe crab blood is valued at \$60,000 per gallon!

Regression Example on Bioanalytical Chemistry



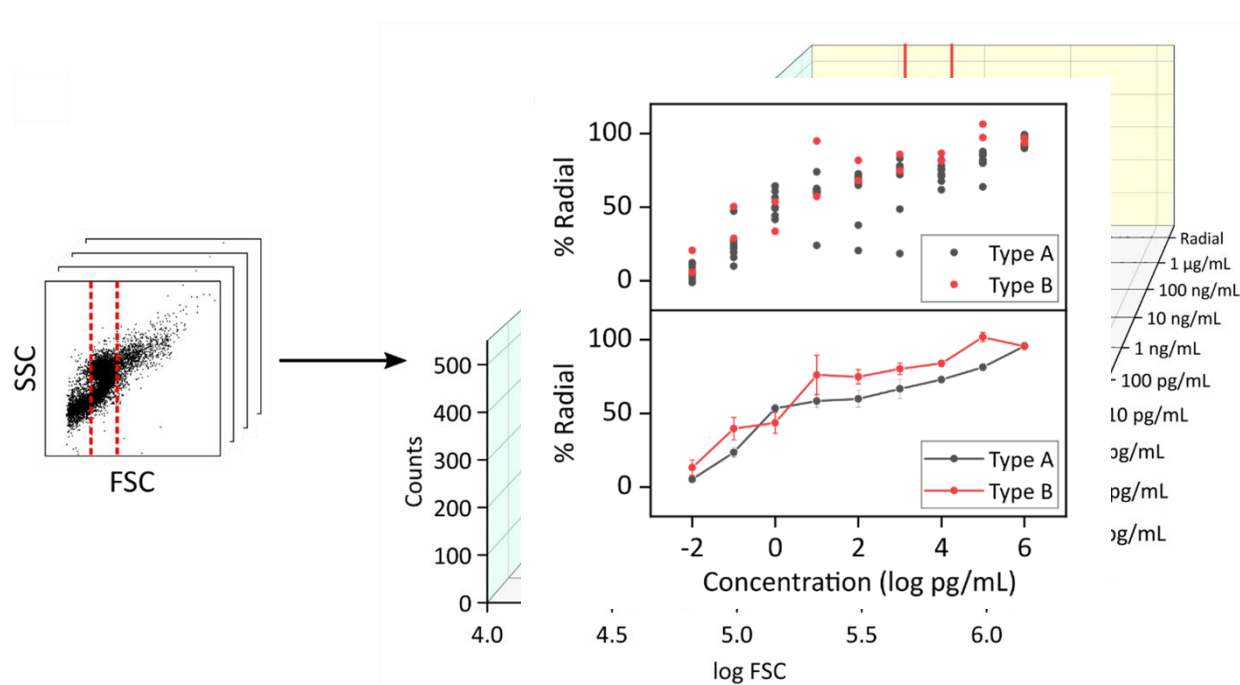
How do we represent flow cytometry scatter plots?

Jiang et al. *Analyst* 146 (2021)

- Endotoxins change the ordering of **liquid crystal** droplets.
- The droplets flow through a laser beam.
- Light scattering is measured at small angles as FSC and large angles as SSC.

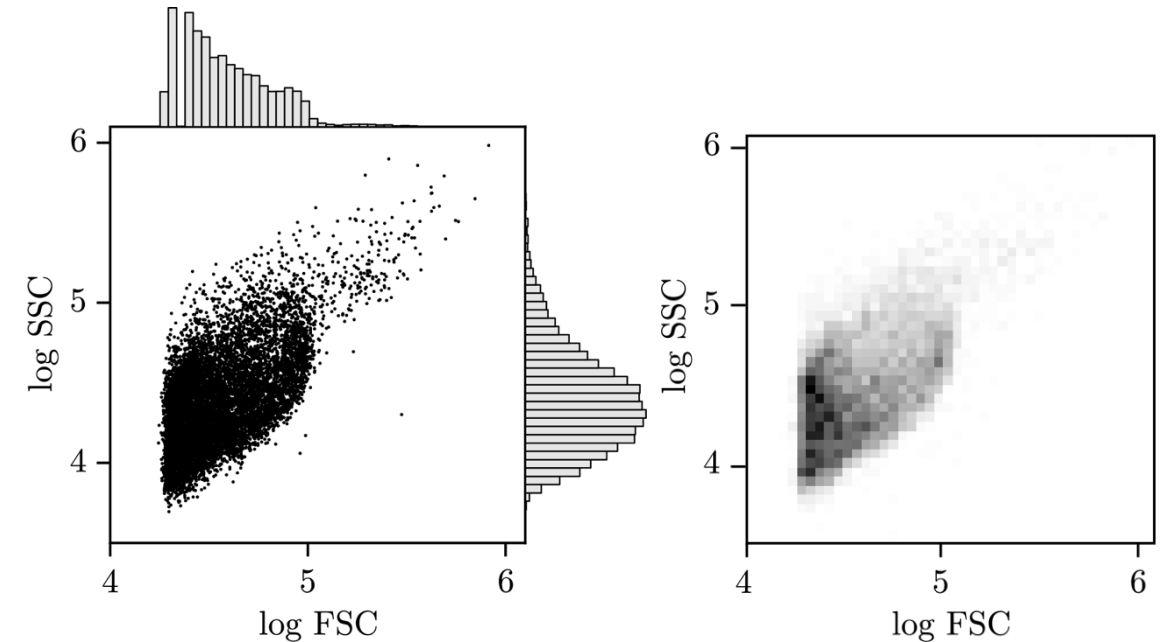
- Flow cytometry is inexpensive and detects endotoxin levels as low as 0.01 pg/mL .
- Scatter patterns change from *bipolar* to *radial* as endotoxin concentration increases.

Regression Example on Bioanalytical Chemistry



Input 1: Radial Configuration Method.

- Counts points between the red lines to produce one scalar, ($x \in \mathbb{R}$).
- This compression loses information and increases uncertainty.

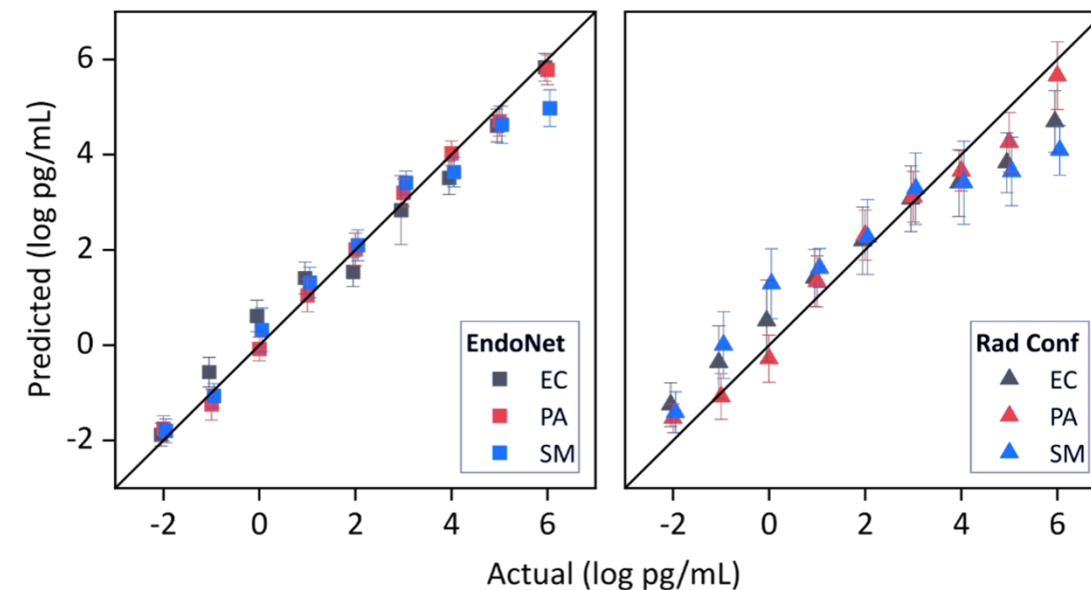
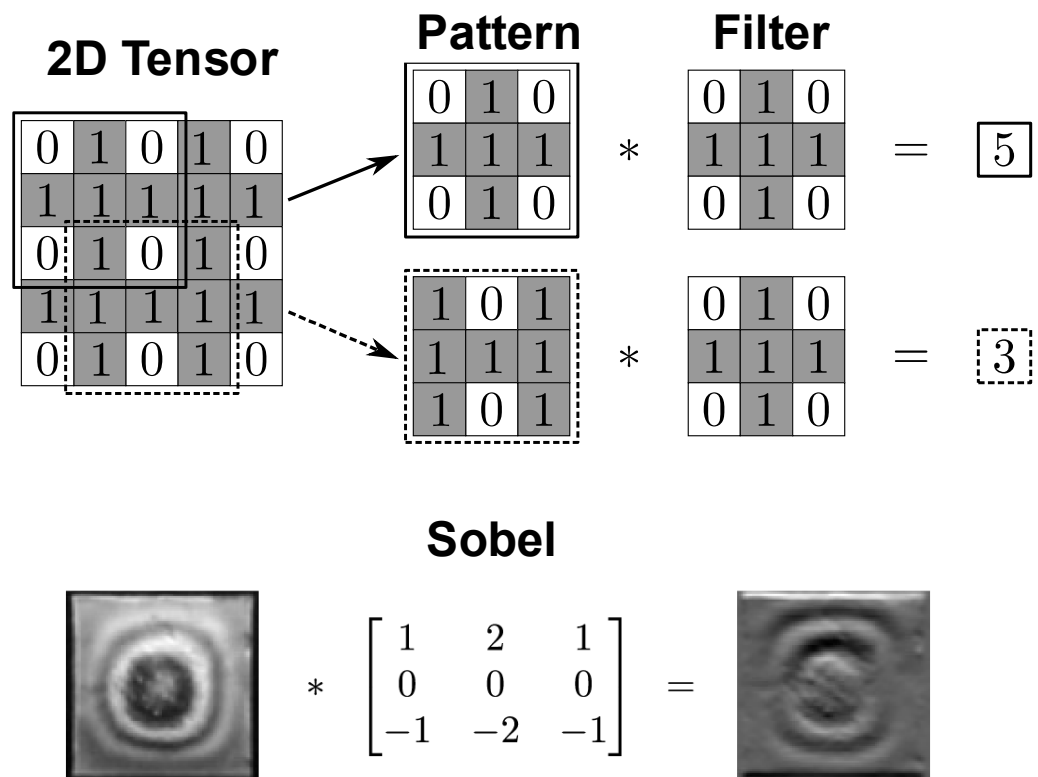


Input 2: Histogram Method.

- Counts points in each 2D bin to create a grayscale matrix ($X \in \mathbb{R}^{m \times n}$).
- The matrix preserves complex spatial patterns.

- **Output:** Endotoxin concentration ($y \in \mathbb{R}$).
- **Task:** Supervised Regression.

Regression Example on Bioanalytical Chemistry

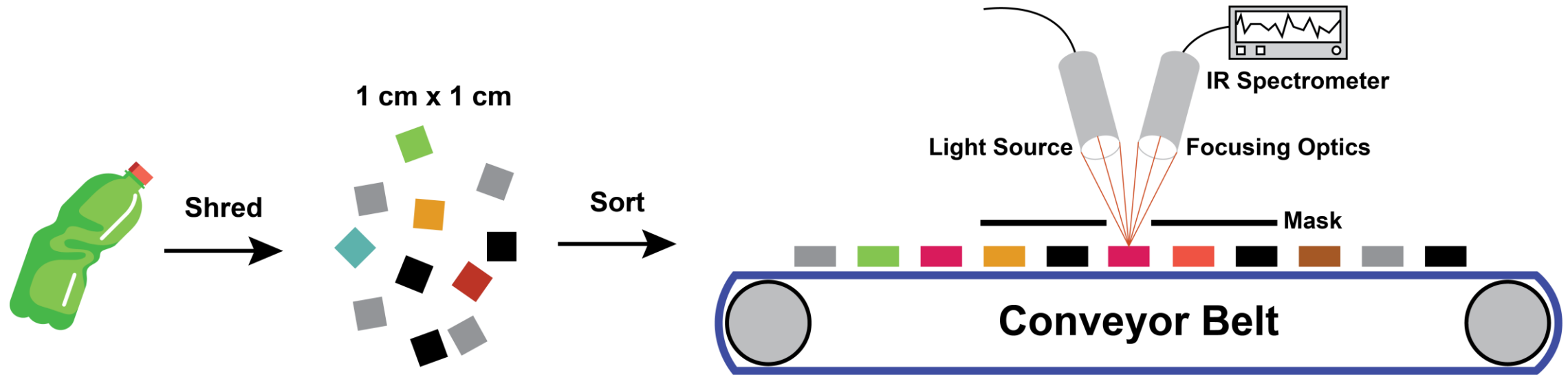


Model: Convolutional Neural Network (CNN).

- Convolutional filters *identify and quantify* patterns in tensors.

By learning complex patterns in flow cytometry scatter plots, the CNN achieves a high prediction accuracy with an $R^2=0.91$, compared to the RC method ($R^2=0.68$).

Classification Example on Spectral Analysis

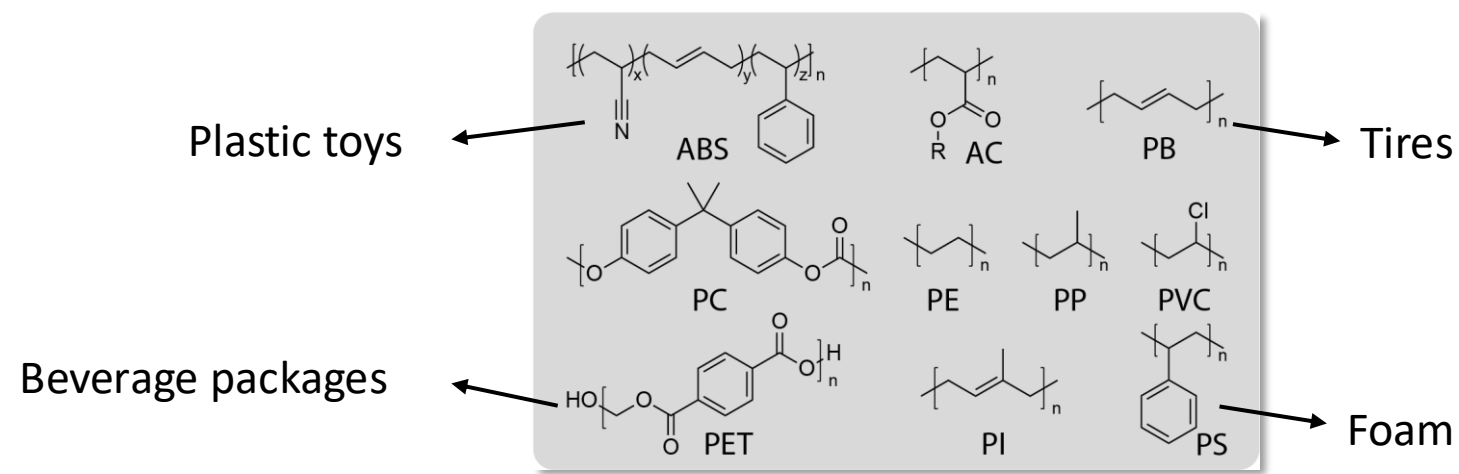


Long et al. *ACS Sustainable Chem. Eng.* 10 (2022)

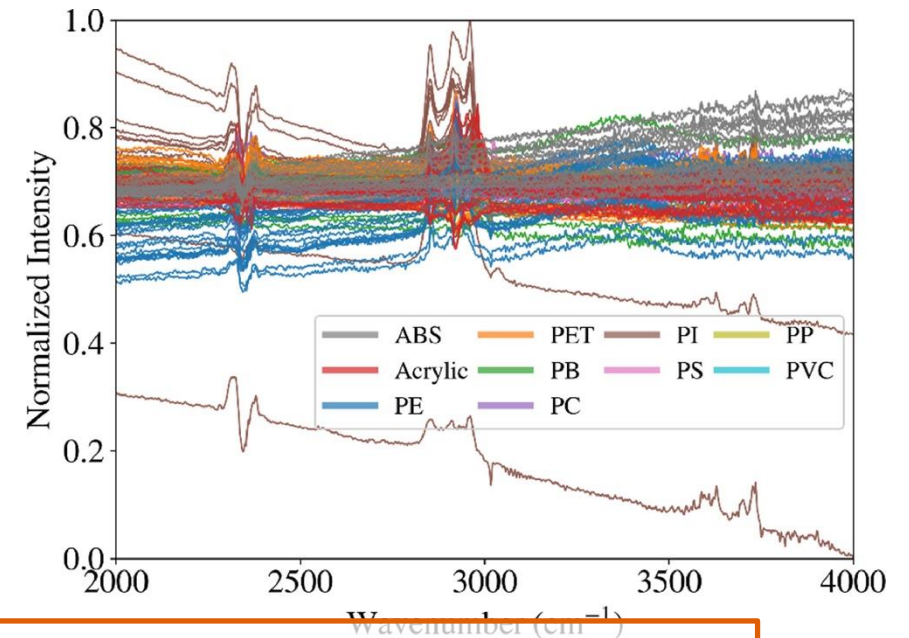
Motivation

- About 6% of all plastic produced through 2015 had been recycled, while about 30% remained in use.
- Mixed plastics are often recycled into low-value products.
- *ATR-FTIR* enables high-throughput sorting by plastic type.

Classification Example on Spectral Analysis



Jiang et al. *Comp. Chem. Eng.* 155 (2021)



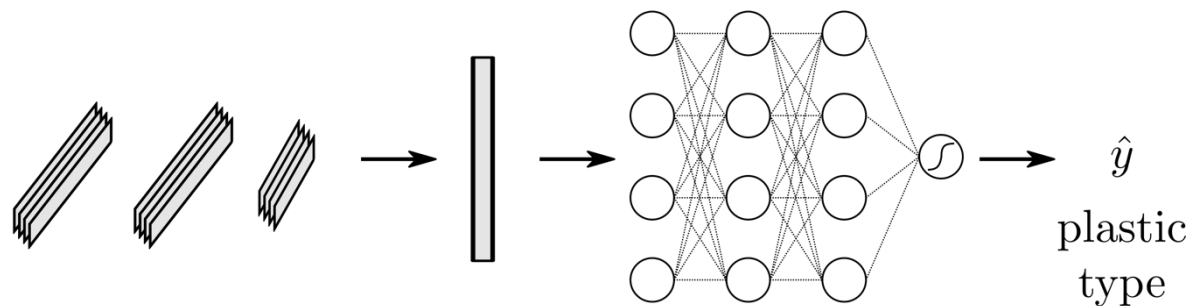
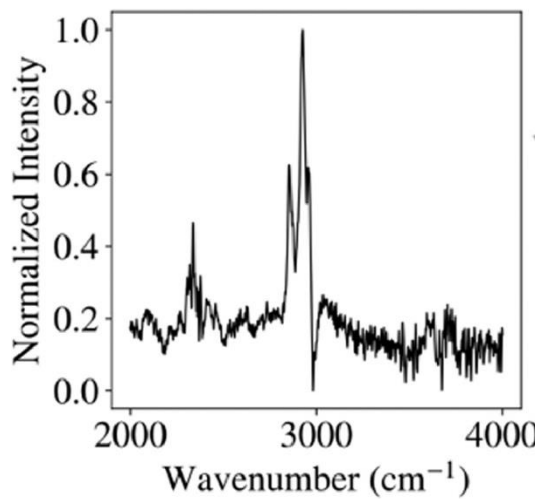
How do we represent ATR-FTIR spectra?

- This study involves 10 plastic types, with 70 spectral measurements for each.
- The spectra exhibit significant overlap between different plastic types, along with noise and systematic errors.

Task: Supervised Classification.

Output: Probability vector indicating the likelihood of each plastic type ($\mathbf{y} \in \mathbb{R}^{10}$).

Classification Example on Spectral Analysis



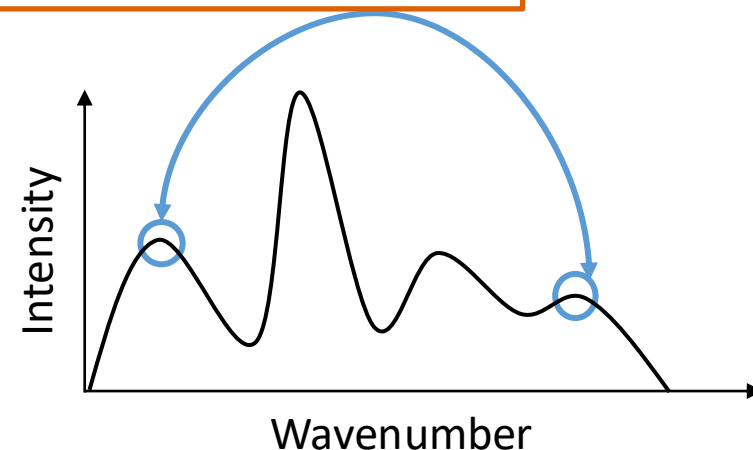
Predicted Species	ABS	AC	PE	PET	PB	PC	PI	PS	PP	PVC
ABS	0.80	0.04	0.01	0.00	0.13	0.03	0.00	0.01	0.00	0.03
AC	0.00	0.61	0.00	0.00	0.04	0.06	0.00	0.01	0.06	0.00
PE	0.00	0.00	0.97	0.00	0.01	0.00	0.00	0.00	0.06	0.00
PET	0.06	0.01	0.00	0.76	0.04	0.00	0.04	0.00	0.00	0.01
PB	0.06	0.04	0.00	0.04	0.46	0.03	0.01	0.00	0.04	0.03
PC	0.06	0.10	0.01	0.00	0.04	0.69	0.00	0.01	0.04	0.04
PI	0.01	0.01	0.00	0.09	0.01	0.00	0.93	0.00	0.00	0.01
PS	0.00	0.06	0.00	0.01	0.04	0.10	0.00	0.91	0.00	0.00
PP	0.00	0.10	0.00	0.00	0.01	0.10	0.01	0.00	0.79	0.01
PVC	0.01	0.01	0.00	0.10	0.20	0.00	0.00	0.04	0.01	0.86

What if we encode long-range correlations directly?

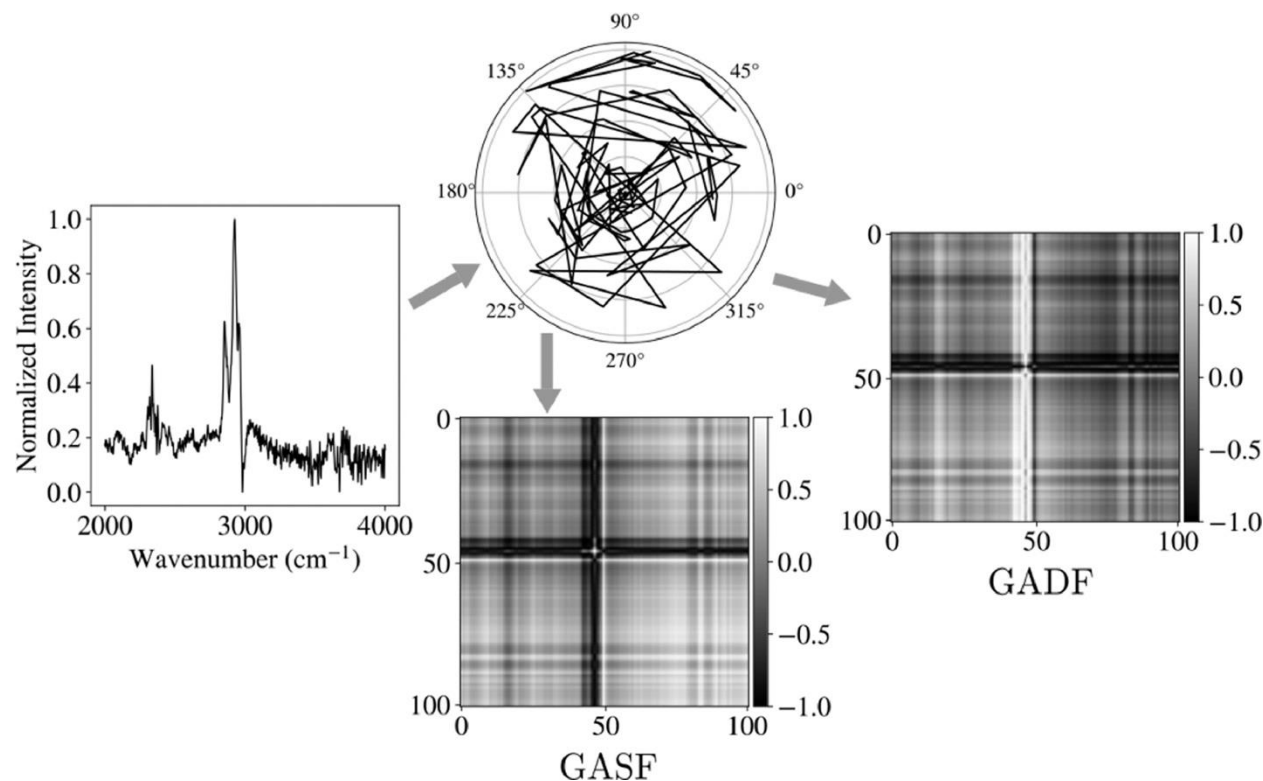
Input 1: Vector (1D Tensor).

Model 1: 1D CNN.

- Each spectrum contains 4,150 intensity values ($\mathbf{x} \in \mathbb{R}^{4150}$).
- The 1D CNN achieves an accuracy of 0.78.
- 1D CNN does not fully capture long-range correlations across wavenumbers.



Classification Example on Spectral Analysis



Input 2: Matrix (2D Tensor).

Model 2: 2D CNN.

- The matrix representation (**Gramian Angular Field, $X \in \mathbb{R}^{100 \times 100}$**) captures correlations across wavenumbers.
- *In this study, changing the representation improves accuracy from 0.78 to 0.87.*

IR Spectrum x

Polar Representation

$$\phi_i = \arccos(x_i)$$
$$r_i = \frac{i}{4150}, i = 1, \dots, 4150$$

$$GASF = \cos(\phi_i + \phi_j)$$
$$GADF = \sin(\phi_i - \phi_j)$$

Dimension Reduction

$$GASF \in \mathbb{R}^{100 \times 100}$$
$$GADF \in \mathbb{R}^{100 \times 100}$$

Classes of Machine Learning

Unsupervised Learning

- In **unsupervised learning**, we aim to create a model that identifies patterns in $\mathbf{x}$.
- The optimization/learning of our model is ***unsupervised*** because the algorithm will not exploit knowledge of labels over the dataset

Unsupervised learning is usually used for

Clustering – Groups similar samples without known labels.

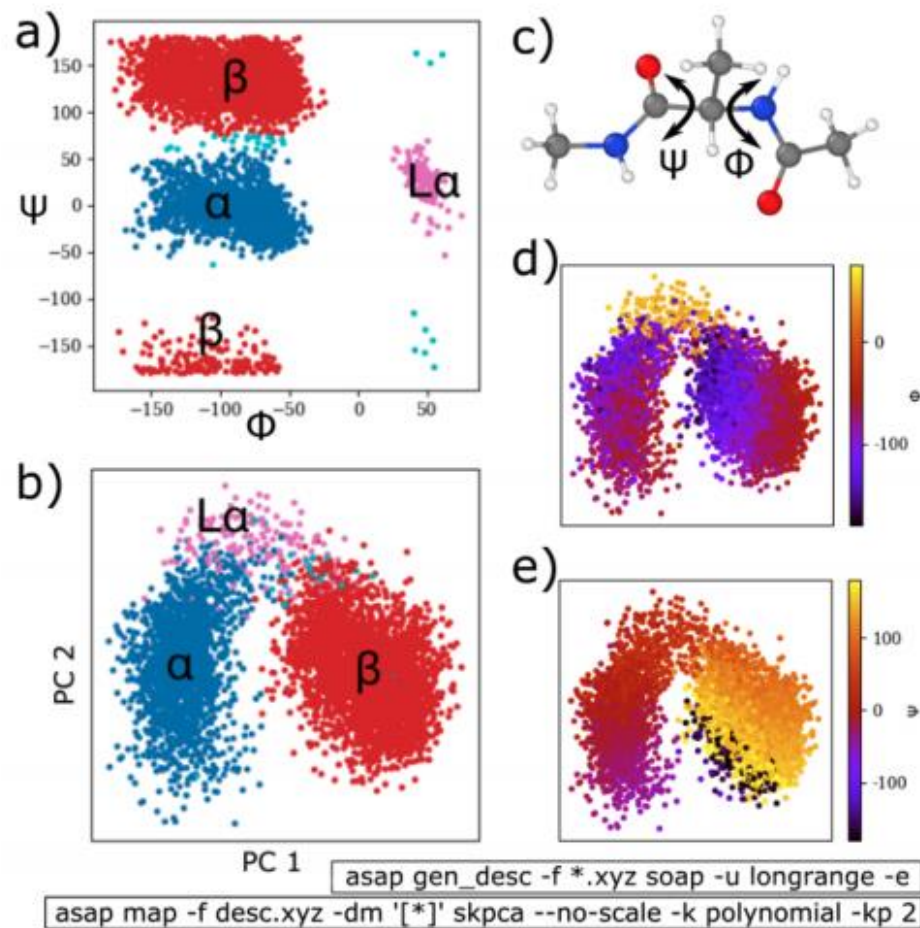
e.g., chemical classes

Dim reduction – Represents high-dimensional data using fewer variables while preserving important structure.

e.g., PCA of molecular configurations

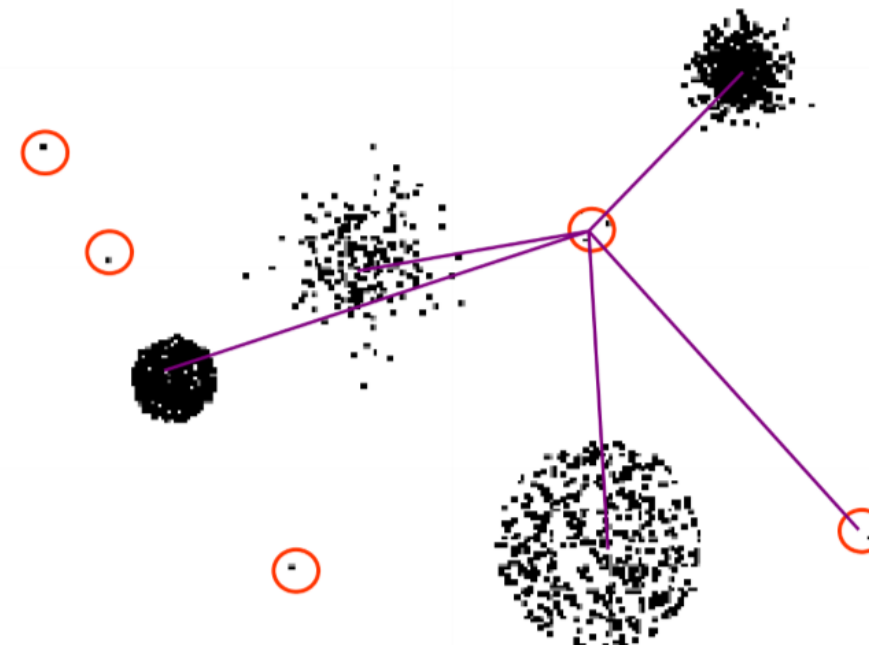
Unsupervised Learning Example

Dimensionality Reduction



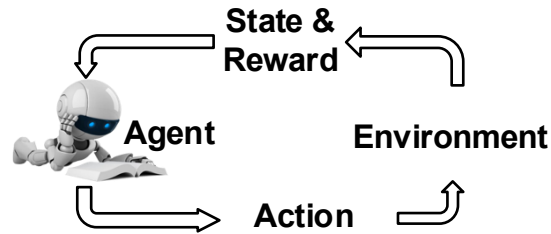
The Ramachandran plot uses the known dihedral angles. PCA reduces high-dimensional molecular descriptors to PC1 and PC2, recovering similar conformational states without using these angles.

Outlier Detection



The model learns the normal structure of the data without labels. Isolated points far from the main clusters are identified as outliers.

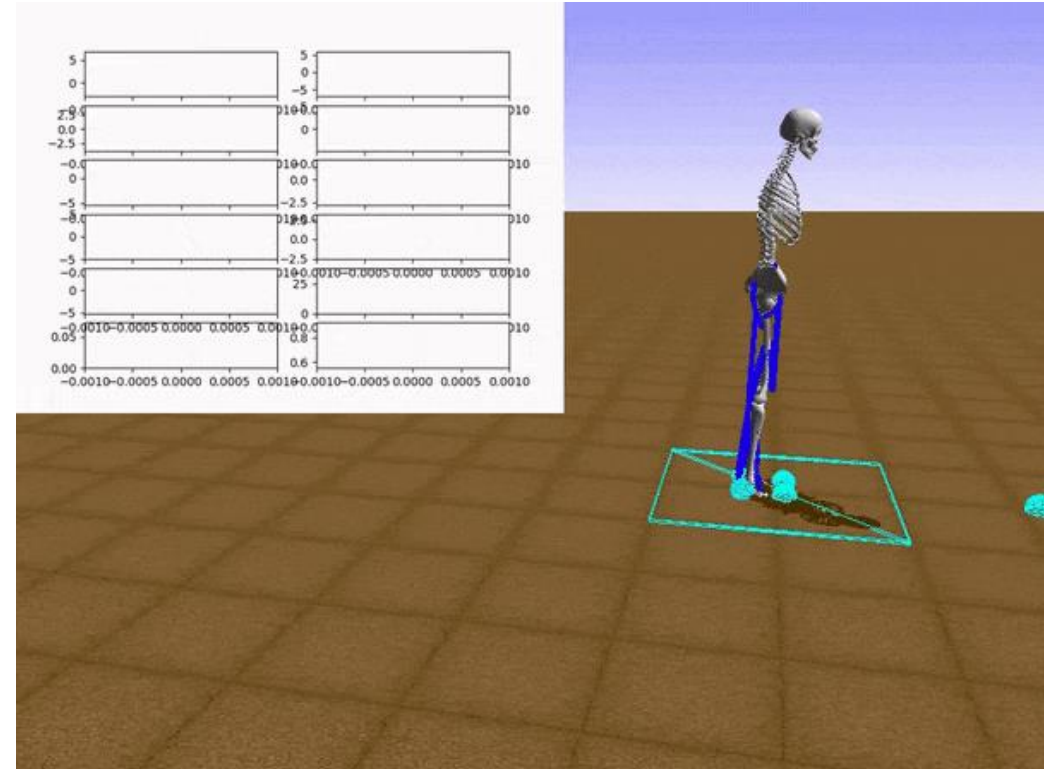
Classes of Machine Learning



Reinforcement Learning

- In reinforcement learning, an “agent” learns how to interact with its environment based on feedback via cumulative rewards/penalties
- The agent learns a policy that maximizes cumulative reward over time.
- Used for sequential decision problems in which actions affect future states and rewards.

e.g. automated process synthesis, process control



Quick Conceptual Quiz

A. Regression

B. Classification

C. Clustering

D. Dimensionality reduction

E. Outlier detection

F. Reinforcement learning

Choose the most appropriate learning task:

1. Flag unusual batches that differ strongly from historical production.
2. Select synthesis conditions through repeated experiments and reward feedback.
3. Determine whether a water sample is safe or contaminated.
4. Represent 200 manufacturing measurements using three variables.
5. Predict the remaining lifetime of a lithium-ion battery.
6. Organize unlabeled drug molecules according to similar properties.